started. The company had been developing reclaimed-water technology since 2015, established a pilot line in 2016, and began building a reclaimed-water plant in the Southern Taiwan Science Park in 2020. The plant became operational in 2022, treating industrial wastewater for reuse in semiconductor manufacturing.

Taiwan showed that even an established semiconductor hub can face a manufacturing risk when its water supply comes under pressure. For the industry, conserving water alone would not be enough. More of it would have to be recovered, treated, and brought back into the manufacturing cycle.

Closing the water loop

Taiwan showed that simply finding more water was not a long-term answer. Chipmakers had to get better at using the water they already had. That meant treating wastewater, recovering it, and putting it back into the manufacturing process instead of using fresh water every time.

TSMC has gradually pushed reclaimed water into more parts of chipmaking. It first used reclaimed water for less critical applications before qualifying it for more demanding processes. In 2024, the company said reclaimed water had been approved for its 5nm and 3nm processes. Its reclamation plants in Taiwan were supplying about 67,000 cubic metres of water a day.

Samsung Semiconductor is taking a similar route. Its facilities treat and reuse wastewater, while recovered water is fed back into operations. In 2024, its Korean sites reused about 101 million tonnes of water, with process improvements saving another 16.76 million tonnes. The company is also working with Korean authorities to bring treated municipal wastewater into its fabs, with plans to supply 120,000 tonnes a day to its Giheung and Hwaseong sites from 2029.

Intel is taking the idea a step further by looking beyond its own fabs. It combines water reuse with projects that restore water in the communities around its operations. The company wants to become net positive for water by 2030 and said it conserved about 42.4 billion litres in 2025, while helping restore another 10.6 billion litres.

The direction is clear. Chipmakers are trying to make the same water go through more cycles before they need fresh supplies. But there is a limit to how much can be recovered. Some water is still lost during treatment, cooling, and other processes. And as fabs keep getting bigger and move into places where water is already scarce, even that remaining demand starts to matter.

India enters the fray

India has watched the semiconductor industry deal with its water problem in Taiwan, South Korea, and elsewhere. Now, as the country starts building fabs of its own, it has a chance to avoid making the same mistake: treating water as an infinite commodity.

Dholera offers an early look at that challenge. Tata Electronics' planned 300mm fab is designed for 50,000 wafers a month, and the project includes systems for ultrapure water and wastewater treatment, with technologies such as reverse osmosis and ion exchange.

These systems may sit behind the scenes of chip production, but they play a critical role in keeping the manufacturing process running. The ability to treat, reuse, and reliably supply water can therefore become an important part of fab operations as production scales.

Micron's Sanand facility, which began commercial production in February 2026, has already given India experience in commercial semiconductor manufacturing, although the facility currently focuses on assembly and test rather than wafer fabrication. Dholera will take the country into the more water-intensive stage of semiconductor production.

The timing matters. India is trying to build a semiconductor industry at a point when water is already becoming a constraint for fabs elsewhere. The question is whether India will build water resilience into its semiconductor infrastructure from the beginning, or learn the hard way that a chip factory needs more than land, power, and cleanrooms.

The next challenge to chip making

Recycling might change the way fabs think about water, but it does not change one basic fact that chips still need fresh water to be made. Some water is lost through treatment, evaporation, and cooling, while newer and more complex processes keep adding to demand. TSMC shows the gap. Its water use per wafer fell, yet total consumption rose, showing how production growth can erase efficiency gains.

That is where the challenge gets harder. Recycling can reduce the amount of fresh water a fab needs, but it cannot keep pace with growth forever. More fabs and more wafers can lead to more water having to be treated, recovered, and replaced. The industry might therefore have to look beyond recycling and find ways to reduce the amount of freshwater needed to make each wafer in the first place.

The industry has spent decades learning how to make smaller transistors, faster chips, and more powerful devices. Its next efficiency challenge may be, *"Can it keep making more chips without taking more freshwater from the world around it?"* EFY

Saba Aafreen, Tech Journalist at EFY, blends on-ground industrial experience with a growing focus on AI-driven technologies in the evolving electronics industry.